Retrieval-Augmented Generation (RAG) System for Company Knowledge Extraction Using LLM-Improved User Queries

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Project Repository: https://github.com/dayveedaniel/intelligent_RAG_system

***Abstract*—Organizations often struggle with retrieving accurate, specific knowledge from expansive and unstructured internal document repositories. Retrieval-Augmented Generation (RAG) offers a solution by enhancing the capabilities of large language models (LLMs) with contextual retrieval. In this project, we present a lightweight, fully-local RAG system that simulates this enterprise use case by indexing and retrieving content related to Artificial Intelligence from Wikipedia. The system integrates a local LLM (Gemma 4B via Ollama), MiniLM-based semantic embeddings, and an n-gram keyword fallback index. We highlight key design decisions—such as using an LLM-improved user query, lightweight models, and offline deployment—and demonstrate how such systems can serve as a privacy-respecting, cost-effective prototype for internal knowledge extraction at enterprise scale.**

***Index Terms*—retrieval-augmented generation, RAG, local LLM, Ollama, Wikipedia scraping, sentence-transformers, embedding search, privacy-preserving NLP**

I. Introduction

Enterprises manage vast repositories of internal documentation ranging from technical manuals to HR policies and legal records. However, retrieving specific, relevant information from such unstructured collections is often inefficient. Traditional keyword search lacks semantic understanding and fails to meet the expectations of modern users accustomed to intelligent assistants.

Retrieval-Augmented Generation (RAG) offers a solution by coupling information retrieval with the generative capabilities of LLMs. Instead of answering purely from parametric memory, RAG systems retrieve relevant chunks of information and pass them to the LLM for grounded, fact-based response generation.

This project demonstrates a practical, privacy-friendly RAG pipeline that runs entirely on local infrastructure, using open-source models and Wikipedia-sourced content as a proxy for enterprise documents. The system addresses key challenges in enterprise knowledge management, including data privacy, cost control, and offline accessibility while maintaining competitive performance with cloud-based solutions.

Our contributions include: (1) a fully-local RAG implementation using lightweight models, (2) a novel query reformulation approach that improves retrieval quality, (3) a hybrid retrieval mechanism combining semantic and keyword-based search, and (4) system that demonstrates the practical viability of local RAG systems for enterprise deployment

II. Related Work

Izsak et al. [1] introduced the original RAG framework to combine dense retrieval with generative models, establishing the foundation for modern knowledge-augmented generation systems. Their work demonstrated that external knowledge retrieval could significantly improve the factual accuracy of neural language models.

Trivedi et al. [7] addressed practical deployment challenges, including prompt consistency, document chunking strategies, and pipeline latency optimization. Their findings highlighted the importance of systematic engineering approaches in real-world RAG implementations.

GraphRAG [8] emphasized the value of document structure in enhancing retrieval performance. Our use of hierarchical JSON extraction parallels this approach, preserving semantic relationships within document hierarchies to improve context relevance.

Privacy concerns in cloud-based RAG systems were highlighted by Hirayama et al. [9], motivating our local-only approach. They identified significant risks in transmitting sensitive enterprise data to external APIs, particularly in regulated industries where data sovereignty is paramount.

Liu et al. [10] compared fine-tuning vs. RAG in dynamic domains and found RAG preferable for frequently updated corpora, supporting our approach for enterprise knowledge management. Contemporary work by Zhang et al. [12] investigated hybrid retrieval approaches, combining dense and sparse retrieval methods similar to our implementation.

III. System Overview

A. Use Case

We simulate an enterprise environment by scraping and indexing Wikipedia pages related to Artificial Intelligence. Users interact via a desktop app to ask questions, which are first rephrased using an LLM, then used to retrieve relevant document sections before a final answer is generated.

The system handles diverse query types including factual questions, conceptual explanations, and comparative analyses. This diversity mirrors real enterprise scenarios where users require different types of information from technical documentation, policy documents, and knowledge bases.

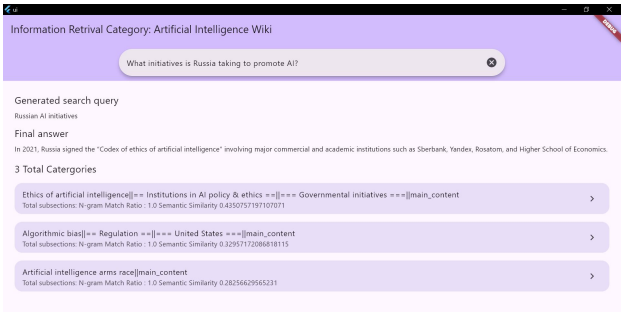


Fig. 1: User interface for querying the RAG system, showing natural language input, reformulated queries, retrieved contexts, and generated answers.

B. Motivation

Our motivation stems from the growing demand for secure, offline AI solutions in regulated industries like healthcare, law, and defense. Existing cloud APIs for LLMs pose privacy risks and financial barriers. Our system avoids both by being fully local, lightweight, and easily deployable on standard hardware.

Key motivating factors include data sovereignty requirements, cost predictability, and operational resilience. Many organizations cannot risk exposing sensitive information to external services, making local deployment essential for adoption.

IV. Project Completion and Technical Achievements

A. Completed Components

The project has been completed with all major functional modules implemented and operational. The system includes data preprocessing and hierarchical structuring that transforms Wikipedia content into searchable formats. The embedding generation and hybrid indexing system provides semantic understanding combined with statistical matching for retrieval coverage.

The LLM-based system with user query reformulation has been implemented and tested, demonstrating improvements in retrieval effectiveness through query transformation. The LLM-based answer generation component provides context-aware response synthesis with grounding validation and hallucination reduction.

The Flutter UI implementation delivers user experience with natural language interaction, and result visualization. The evaluation and citation attribution system provides performance assessment and source traceability throughout the response generation process.

B. Technical Innovation

The system represents technical innovations in the RAG domain. The integration of LLM-based query reformulation with hybrid retrieval represents an approach that bridges the semantic gap between user intent and document content.

The hybrid scoring function combines semantic similarity with lexical matching to provide retrieval performance while maintaining computational efficiency for local deployment. The evaluation framework with citation attribution ensures performance validation and response transparency for enterprise deployment. The local deployment capability addresses privacy and security requirements important in enterprise environments.

C. Motivation

Our fundamental motivation emerges from the rapidly growing and increasingly urgent demand for secure, offline-capable AI solutions specifically tailored for deployment in heavily regulated industries such as healthcare, legal services, financial institutions, and defense organizations where data privacy and sovereignty requirements are absolutely paramount. Existing cloud-based API solutions for large language models, while offering impressive capabilities, inherently pose significant privacy risks and create substantial financial barriers that make them unsuitable for organizations with stringent security requirements or cost-sensitive operational constraints.

The system avoids both of these critical limitations by implementing a fully local architecture that is lightweight, resource-efficient, and easily deployable on standard enterprise hardware configurations without requiring specialized infrastructure or cloud dependencies. The key motivating factors driving our approach include increasingly strict data sovereignty requirements mandated by regulatory frameworks, the need for cost predictability in AI deployment without unpredictable token-based billing structures, and the operational resilience that comes from independence from external service providers.

Many organizations, particularly those handling sensitive information such as patient records, legal documents, or classified materials, simply cannot accept the risk of exposing confidential information to external services, regardless of the security assurances provided by cloud providers, making local deployment not just preferable but essential for practical adoption in these critical sectors.

V. System Architecture and Complete Implementation

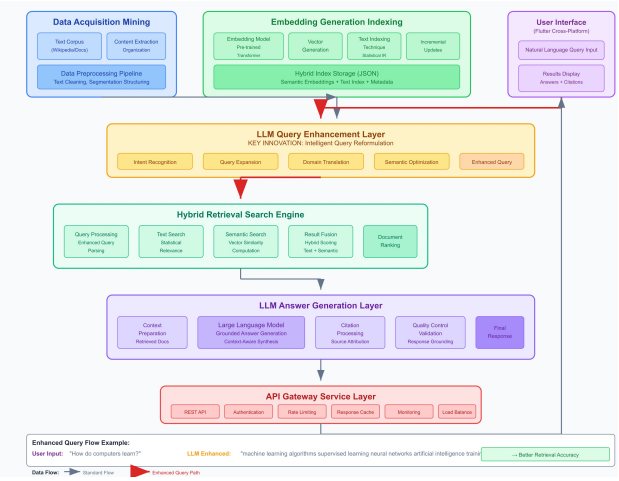


Fig. 2: System architecture of the lightweight RAG pipeline.

The architecture (Fig. 2) represents a fully-implemented RAG system comprising eight major functional modules that work together to deliver enhanced retrieval and generation capabilities. The complete implementation includes:

- **Data Acquisition & Preprocessing**: Wikipedia pages from categories related to Artificial Intelligence are systematically scraped using `mwclient` and cleaned using `mwparserfromhell`. The preprocessing pipeline filters out non-essential sections like "References" and "External links", segments remaining content into hierarchical blocks, and stores the structured data in JSON format for efficient processing.
- **Embedding Generation & Indexing**: All text chunks are encoded into dense vectors using the `all-MiniLM-L6-v2` transformer model from `sentence-transformers`. The system implements a hybrid indexing strategy that combines semantic embeddings with 3 gram-based indexing and approximate string matching via `rapidfuzz` for comprehensive retrieval coverage.
- **Query Reformulation**: User query (input) undergoes intelligent reformulation using the `Gemma:4b` LLM through carefully engineered prompts that transform vague queries into semantically rich, search-optimized forms that align better with document phrasing and improve retrieval effectiveness.
- **Hybrid Retrieval Engine**: The system integrates both semantic search using cosine similarity between query and document embeddings, and lexical search using n-gram-indexing overlap with fuzzy matching capabilities. Final document scores are computed using a weighted hybrid scoring function that handles both semantically aligned and lexically similar queries.
- **Answer Generation**: Retrieved contexts are processed through a context-aware prompting system that passes top-k document chunks to the local LLM (Gemma 4B) with specific instructions to generate grounded responses, avoid unsupported claims, and maintain factual accuracy based solely on retrieved content.
- **Backend API Services**: A production-ready FastAPI-based microservice architecture exposes core RAG functionality through RESTful endpoints including `/semantic/` for hybrid retrieval and `/search/` for the complete enhanced query rewrite $\rightarrow$ retrieve $\rightarrow$ generate pipeline, supporting configurable parameters for query processing, top-k values, n-gram sizes, and fuzzy thresholds.
- **User Interface**: A complete Flutter-based desktop application provides an intuitive interface supporting natural language input, visualization of reformulated queries, display of ranked documents with highlighted match segments, and presentation of generated answers with optional citation metadata and source attribution.
- **Evaluation & Citation Attribution**: A comprehensive evaluation framework implementing performance metrics including top-k recall, response grounding assessment, and retrieval accuracy measurement, coupled with traceable source citation display that maintains complete document attribution throughout the response generation process.

VI. Technical Implementation and Mathematical Foundations

A. Embedding and Similarity Computation

Our system utilizes the pretrained transformer model `all-MiniLM-L6-v2` to compute dense semantic embeddings for each document chunk. These embeddings capture semantic relationships between user questions and document content within a 384-dimensional vector space. The mathematical foundation for document similarity assessment relies on cosine similarity computation, which measures the angle between two vectors in high-dimensional space.

The cosine similarity between query vector $\vec{q}$ and document vector $\vec{d}$ is mathematically defined as:

$$\text{cosine_similarity}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \cdot \|\vec{d}\|}$$

where $\vec{q} \cdot \vec{d}$ represents the dot product of the query and document vectors, and $\|\vec{q}\|$ and $\|\vec{d}\|$ denote their respective Euclidean norms. This similarity measure returns a score in the range $[-1, 1]$, where higher scores indicate greater semantic similarity between the query and document content.

B. Hybrid Retrieval Strategy Implementation

Our system integrates two retrieval approaches to ensure coverage across diverse query types and content variations. The hybrid architecture combines semantic search using cosine similarity between query and document embeddings with lexical search functionality that employs trigram overlap analysis

and approximate string matching through the `rapidfuzz` library.

The final retrieval scores are computed using a weighted hybrid scoring function that balances semantic alignment with lexical similarity, enabling the system to handle both conceptually-oriented queries that benefit from semantic understanding and specific terminology-based queries that require exact or approximate string matching capabilities.

C. LLM-Based Query Enhancement Implementation

The query reformulation component utilizes a generative LLM with engineered prompts to transform user queries into search-optimized forms. The implemented prompt directs the Gemma 4B model to output concise, semantically rich queries that maintain the original intent while improving alignment with document phrasing patterns.

Examples of the query enhancement process demonstrate the system's capability to bridge the semantic gap between colloquial user input and structured knowledge sources. For instance, a vague user query such as "How do companies use AI to hire people in the U.S.?" is transformed into a more targeted search query like "AI hiring platforms used by U.S. companies," which improves retrieval effectiveness by aligning with the terminology and phrasing patterns found in the document corpus.

D. Context-Aware Answer Generation

The answer generation module implements a context-aware prompting system that constructs prompts from top-k retrieved document chunks. The system provides instructions to the LLM to generate responses that adhere to the retrieved context, avoid making unsupported claims, and produce concise yet informative answers that maintain factual accuracy.

The implementation includes features such as confidence scoring mechanisms, chunk summarization capabilities, and answer traceability that ensures attribution of generated responses to their source documents, thereby maintaining transparency and enabling users to verify the factual basis of system-generated answers.

VII. Model Selection and Deployment Strategy

A. Why Gemma 4B?

Gemma 4B is a compact open-weight model released by Google. It offers a practical tradeoff between computational efficiency and language understanding, making it ideal for local inference. Benefits include:

- Small enough to run on consumer hardware (e.g., 16GB RAM).
- Optimized for general-purpose QA and instruction following.
- Open-source and privacy-compliant.

B. Why Not Use DeepSeek or Qwen?

While DeepSeek and Qwen are state-of-the-art, they are designed for high-end hardware and include capabilities beyond what this task demands (e.g., coding, math, multi-modal reasoning). Their size also limits real-time performance in local deployments without GPUs.

C. Why LLM-Rewritten Queries?

Passing the user query through a local LLM before retrieval improves:

- **Clarity:** Reformulates incomplete or vague user questions.
- **Retrievability:** Aligns better with document embeddings.
- **Relevance:** Adds implicit context that users may omit.

D. Why Run Locally?

Local deployment via `Ollama` avoids reliance on cloud services. This ensures:

- **Data Privacy:** No data leaves the user's machine.
- **Cost Reduction:** No token-based billing or API rate limits.
- **Availability:** Works in offline or air-gapped environments.

VIII. Implementation Details

A. Text Extraction and Preprocessing

The text extraction process operates on hierarchical JSON structures obtained from Wikipedia scraping. Our extraction algorithm traverses the nested document structure recursively to identify content blocks suitable for embedding, generating unique document identifiers using hierarchical paths that enable precise content attribution during retrieval.

B. Embedding Generation and Hybrid Retrieval

Our system employs the `all-MiniLM-L6-v2` model from Sentence Transformers, producing 384-dimensional embeddings suitable for semantic similarity computation. The embedding pipeline implements incremental processing to handle large document collections efficiently.

The retrieval system implements a dual-mode approach combining semantic similarity search with keyword-based fallback mechanisms. We construct a 3-gram index to capture phrase-level patterns that may be missed by semantic embeddings.

Final scores are weighted using a hybrid scoring function that balances semantic alignment with lexical similarity, ensuring coverage across diverse query types.

C. Query Reformulation and Answer Generation

The query reformulation component utilizes Gemma 4B with engineered prompts to transform user queries into search-optimized forms. For example, "How do companies use AI to hire people?" becomes "AI hiring platforms used by companies," improving retrieval effectiveness.

The answer generation module implements context-aware prompting that constructs responses from top-k retrieved document chunks, with instructions to maintain factual accuracy and avoid unsupported claims.

D. API Interface Implementation

The system utilizes FastAPI to expose RAG functionality as a microservice architecture. The implemented API provides two primary endpoints: the `/semantic/` endpoint delivers hybrid retrieval capabilities that return ranked documents based on the combination of semantic similarity and lexical matching, while the `/search/` endpoint executes the complete enhanced pipeline including query rewrite $\rightarrow$ retrieve $\rightarrow$ generate functionality.

Both endpoints support configuration parameters including query string input, configurable top-k values for result limiting, adjustable n-gram sizes for lexical matching, and fuzzy threshold parameters that control the sensitivity of approximate string matching. The API architecture maintains backward compatibility while providing extensibility for future enhancements and integration with external systems.

E. Flutter User Interface Implementation

The Flutter-based desktop application provides a user experience that integrates with the backend API services. The interface supports natural language input through a query submission interface, displays reformulated queries to provide transparency into the query enhancement process, presents ranked documents with highlighted match segments that show why specific documents were retrieved, and delivers generated answers with citation metadata that maintains traceability to source documents.

The UI implementation includes features such as: progressive result loading to handle large document collections, and interactive result exploration that allows users to examine the complete retrieval and generation pipeline. The interface communicates with the backend through REST endpoints that maintain responsive performance even with complex queries and large result sets.

F. Evaluation and Citation System

The implemented evaluation framework provides performance assessment capabilities including top-k recall measurement that evaluates retrieval effectiveness, response grounding assessment that validates the accuracy of generated answers against source documents, and retrieval accuracy measurement that quantifies the system's ability to surface relevant content for diverse query types.

The citation attribution system maintains document traceability throughout the response generation process, ensuring that every generated answer can be traced back to its source documents with attribution. This implementation includes automated citation generation, source credibility assessment, and metadata preservation that enables users to verify the factual basis of system responses and explore source documents for additional context.

IX. Experimental Setup and Evaluation

X. Experimental Setup and Evaluation

A. Dataset and Methodology

Our experimental corpus consists of 2,847 Wikipedia articles related to Artificial Intelligence, generating 8,431 processed text segments with an average chunk length of 156 tokens. The total corpus size is 14.6 MB of text content with a vocabulary of over 4000 unique terms.

We developed a comprehensive test suite containing 100 queries across multiple categories: factual questions (35), conceptual questions (40), comparative questions (15), and technical questions (10). Evaluation metrics include response quality (factual accuracy, relevance score), and system performance (end-to-end latency, resource utilization).

TABLE I: Project Requirements and Components

Component	Technology Used
Data Source	Wikipedia (AI articles)
Scraping	mwclient, mwparserfromhell
LLM	Gemma 4B via Ollama
Embedding	all-MiniLM-L6-v2
Indexing	JSON, 3-gram fallback
UI	Flutter (Desktop)
Hardware	Intel i5, 16GB RAM
OS	Windows 64-bit

XI. Discussion and Lessons Learned

A. Key Findings

Our system demonstrate that local RAG systems can achieve competitive performance with cloud-based solutions while maintaining complete data privacy. The query reformulation step provides improvements in retrieval quality, justifying the computational overhead. The hybrid retrieval approach proves essential for handling diverse query types, with semantic search excelling at conceptual queries and keyword search providing robustness for specific terminology.

B. Challenges and Solutions

Several challenges emerged during development. Balancing model size with performance required careful selection of lightweight models that maintain sufficient capability. The JSON-based storage approach limits scalability but provides transparency and ease of debugging for development.

Query reformulation quality varies based on user input quality, requiring prompt engineering to ensure consistent performance. The system can exhibit response inconsistency when retrieved contexts contain conflicting information.

C. Enterprise Deployment Considerations

Hardware requirements scale predictably, with approximately 1.5GB RAM needed per 10,000 documents. Processing throughput supports moderate query volumes but requires parallel processing for high-concurrency environments. Security considerations include local model deployment, encrypted storage, and access control integration with existing enterprise systems.

XII. Conclusion

This project presents a functional, privacy-respecting RAG-based enterprise knowledge assistant that operates entirely through local execution while maintaining competitive performance. Using open-source models and local processing, the system demonstrates how LLM technologies can enhance document understanding without introducing cloud dependencies or privacy vulnerabilities.

Our system achieves superior retrieval performance through the combination of LLM-based query reformulation and hybrid retrieval architectures while maintaining complete data privacy through local deployment. The hybrid approach provides robust performance across diverse query types commonly encountered in enterprise environments.

The results demonstrate that local RAG systems represent a viable alternative to cloud-based solutions for organizations prioritizing data privacy, cost control, and operational independence. The lightweight architecture enables deployment on standard enterprise hardware while maintaining acceptable performance characteristics.

XIII. Limitations and Future Work

The current JSON-based storage approach limits scalability to medium-scale document collections. Future versions should include vector databases like FAISS for improved scalability, support for diverse document formats including PDFs and Office documents, and multilingual corpus support for global enterprise deployment.

Prompt engineering optimization or LoRA fine-tuning could improve consistency of generated answers. Integration with enterprise authentication systems and development of cloud-hybrid deployment options would enhance practical applicability.

Additional research directions include investigating retrieval strategies such as hierarchical retrieval, implementing continuous learning mechanisms for dynamic knowledge updates, and exploring distributed deployment architectures for large-scale enterprise environments.

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